SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2527–2551; original printed pp. 84–85; physical source-PDF pp. 75–76; recomposed running pp. 67–68. This bounded unit is translated and independently source-reviewed against the French TeX and direct PDF. Cumulative integration remains pending. Line 2552 is blank; the next lemma begins at line 2553 and is excluded.

Of course, 2) follows from 1) by observing that, if M and N are two finitely generated A-modules and if we put $\mathcal{F} = \widetilde{M}$ and $\mathcal{G} = \widetilde{N}$, then there are isomorphisms

$$\mathcal{H}_Y(\mathcal{F}) \simeq \widetilde{\mathrm{H}_Y(X,\mathcal{F})},$$
$$\mathcal{E}xt_Y(\mathcal{F},\mathcal{G}) \simeq \widetilde{\mathrm{Ext}_Y(\mathcal{F},\mathcal{G})},$$
$$\mathcal{E}xt_{\mathcal{O}_X}(\mathcal{F},\mathcal{G}) \simeq \widetilde{\mathrm{Ext}_A(M,N)}.$$

Let $\mathcal{C}$ be the category of A-modules, and let **Ab** be the category of abelian groups. Let $\underline{\mathrm{F}}$ be the functor

$$\underline{\mathrm{F}}\colon \mathcal{C} \longrightarrow \mathbf{Ab},$$
$$M \longmapsto \Gamma_Y(\widetilde{M}).$$

We know from Exposé II that there is an isomorphism of ∂-functors

$$\mathrm{H}^*_Y(X,\widetilde{M}) \simeq \mathrm{R}^*\underline{\mathrm{F}}(M).$$

Furthermore, let Ext^*_Y denote the right derived functors, in the second argument, of

$$\underline{\mathrm{F}} \circ \mathrm{Hom}\colon \mathcal{C}^\circ \times \mathcal{C} \longrightarrow \mathbf{Ab}.$$

We know from Exposé VI that there is an isomorphism of ∂-functors

$$\mathrm{Ext}^*_Y(M,N) \simeq \mathrm{Ext}^*_Y(\widetilde{M},\widetilde{N}).$$

Finally, let us record the following result from Exposé VI: if C is an injective A-module and N is a finitely generated A-module, the sheaf

$$\mathcal{H}om(\widetilde{N},\widetilde{C}) \simeq \widetilde{\mathrm{Hom}(N,C)}$$

is flasque; hence

$$\mathrm{R}^1\underline{\mathrm{F}}(\mathrm{Hom}(N,C)) = 0.$$